

DATA ANALYTICS WITH COGNOS - GROUP 5

PROJECT: CUSTOMER CHURN PREDICTION

PHASE 2: INNOVATION

TEAM MEMBER

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PHASE 2 - DOCUMENT SUBMISSION



INTRODUCTION

Welcome to the Customer Churn Prediction project! In today's competitive business landscape, retaining customers is paramount. This project is dedicated to developing a robust and data-driven approach to foresee customer churn, allowing us to take proactive measures to retain valuable clientele.

Customer churn, or the loss of customers, can have a significant impact on a company's bottom line. By harnessing the power of data analytics, machine learning, and predictive modeling, our goal is to predict when and why customers might leave our services, enabling us to implement targeted retention strategies.

Throughout this project, we will delve into data analysis, feature engineering, model selection, and evaluation techniques to build a reliable churn prediction system. Our ultimate objective is to not only identify potential churners but also understand the underlying factors driving their decisions.

Join us on this journey to reduce customer churn, enhance customer satisfaction, and secure the future success of our business. Together, we'll unlock insights from data that will empower us to build stronger and more lasting customer relationships.

The innovation phase of the customer churn prediction project marks a pivotal moment in our journey to enhance customer retention and satisfaction. In this phase, we will explore cutting-edge technologies, data analysis techniques, and predictive models to revolutionize how we identify and mitigate customer churn. By harnessing the power of innovation, we aim to create solutions that not only anticipate churn but also provide actionable insights for proactive customer engagement. Join us on this exciting quest to redefine customer relationship management and secure the future success of our business.

CONTENT FOR PROJECT PHASE 2

Consider incorporating advanced machine learning techniques, such as ensemble models or feature engineering, to improve prediction accuracy.

DATA SOURCE

Telco customer churn:

The dataset contains 7043 rows and 21 features. Each row represents a customer, each column contains the customer's attributes described in the column Metadata

Dataset Link: <https://www.kaggle.com/datasets/blastchar/telco-customer-churn>

customerID	gender	SeniorCiti	Partner	Depender	tenure	PhoneSer	MultipleLi	InternetS	OnlineSec	OnlineBac	DevicePrc	TechSupp	Streaming	Streaming	Contract	Paperless	PaymentM	MonthlyC	TotalChar	Churn
7590-VHV	Female	0	Yes	No		1	No	No phone DSL	No	Yes	No	No	No	No	Month-to	Yes	Electronic	29.85	29.85	No
5575-GNV	Male	0	No	No		34	Yes	No	DSL	Yes	No	Yes	No	No	One year	No	Mailed ch	56.95	1889.5	No
3668-QPYI	Male	0	No	No		2	Yes	No	DSL	Yes	Yes	No	No	No	Month-to	Yes	Mailed ch	53.85	108.15	Yes
7795-CFOI	Male	0	No	No		45	No	No phone DSL	Yes	No	Yes	Yes	No	No	One year	No	Bank tran	42.3	1840.75	No
9237-HQIT	Female	0	No	No		2	Yes	No	Fiber opti	No	No	No	No	No	Month-to	Yes	Electronic	70.7	151.65	Yes
9305-CDSh	Female	0	No	No		8	Yes	Yes	Fiber opti	No	No	Yes	No	Yes	Month-to	Yes	Electronic	99.65	820.5	Yes
1452-KIOV	Male	0	No	Yes		22	Yes	Yes	Fiber opti	No	Yes	No	No	Yes	Month-to	Yes	Credit car	89.1	1949.4	No
6713-OKO	Female	0	No	No		10	No	No phone DSL	Yes	No	No	No	No	No	Month-to	No	Mailed ch	29.75	301.9	No
7892-POO	Female	0	Yes	No		28	Yes	Yes	Fiber opti	No	No	Yes	Yes	Yes	Month-to	Yes	Electronic	104.8	3046.05	Yes
6388-TABK	Male	0	No	Yes		62	Yes	No	DSL	Yes	Yes	No	No	No	One year	No	Bank tran	56.15	3487.95	No
9763-GRSh	Male	0	Yes	Yes		13	Yes	No	DSL	Yes	No	No	No	No	Month-to	Yes	Mailed ch	49.95	587.45	No
7469-LKBC	Male	0	No	No		16	Yes	No	No	No intern	No intern	No intern	No intern	No intern	Two year	No	Credit car	18.95	326.8	No
8091-TTVZ	Male	0	Yes	No		58	Yes	Yes	Fiber opti	No	No	Yes	No	Yes	One year	No	Credit car	100.35	5681.1	No
0280-XJGE	Male	0	No	No		49	Yes	Yes	Fiber opti	No	Yes	Yes	No	Yes	Month-to	Yes	Bank tran	103.7	5036.3	Yes
5129-JLPI	Male	0	No	No		25	Yes	No	Fiber opti	Yes	No	Yes	Yes	Yes	Month-to	Yes	Electronic	105.5	2686.05	No
3655-SNQ	Female	0	Yes	Yes		69	Yes	Yes	Fiber opti	Yes	Yes	Yes	Yes	Yes	Two year	No	Credit car	113.25	7895.15	No
8191-XVWS	Female	0	No	No		52	Yes	No	No	No intern	No intern	No intern	No intern	No intern	One year	No	Mailed ch	20.65	1022.95	No
9959-WOF	Male	0	No	Yes		71	Yes	Yes	Fiber opti	Yes	No	Yes	No	Yes	Two year	No	Bank tran	106.7	7382.25	No
4190-MFLI	Female	0	Yes	Yes		10	Yes	No	DSL	No	No	Yes	Yes	No	Month-to	No	Credit car	55.2	528.35	Yes
4183-MYFI	Female	0	No	No		21	Yes	No	Fiber opti	No	Yes	Yes	No	No	Month-to	Yes	Electronic	90.05	1862.9	No

DATA COLLECTION AND PREPROCESSING

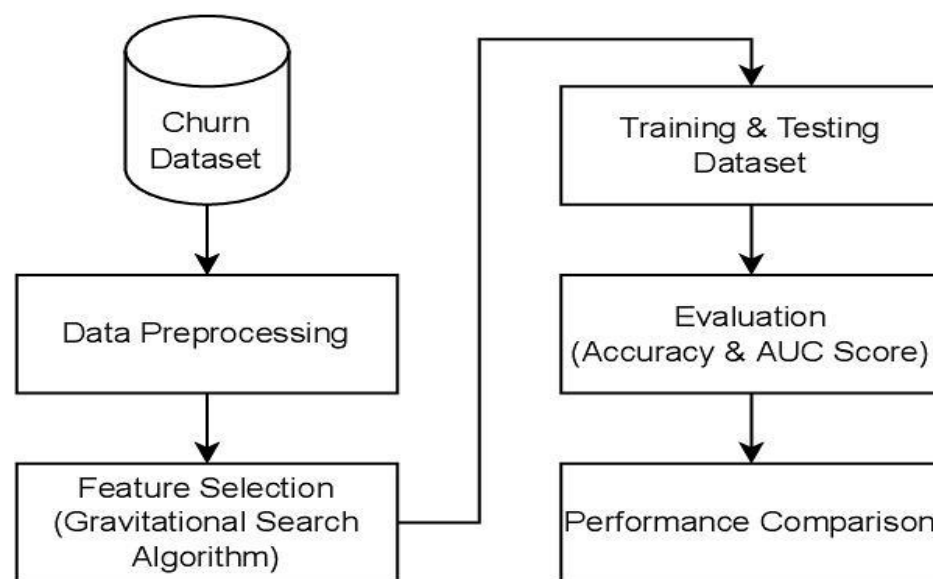
Importing dataset: The dataset contains 7043 rows and 21 features. Each row represents a customer, each column contains the customer's attributes described in the column Metadata. The data set includes information about:

- Customers who left within the last month — the column is called Churn — this is the target feature.
- Services that each customer has signed up for — phone, multiple lines, internet, online.security, online backup, device protection, tech support, and streaming TV and movies.
- Customer account information — how long they've been a customer, contract, payment method, paperless billing, monthly charges, and total charges.
- Demographic info about customers — gender, age range, and if they have partners and dependents.

Data preprocessing: The most common churn prediction models are based on older statistical and data-mining methods, such as logistic regression and other binary modeling techniques.

Raw historical data for solving the problem and building predictive models needs to be transformed into a format suitable for machine learning algorithms. This step can also improve overall results by increasing the quality of data.

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EXPLORATORY DATA ANALYSIS (EDA)

Customer Churn means whether the customer has stopped using the product and switched to a competitor product. To prevent customers from churning, we can use data analysis to identify potential features and design strategies to retain customers from leaving. It is also noted that an already loyal customer is less likely to churn and easier to retain than trying to gain a new customer.

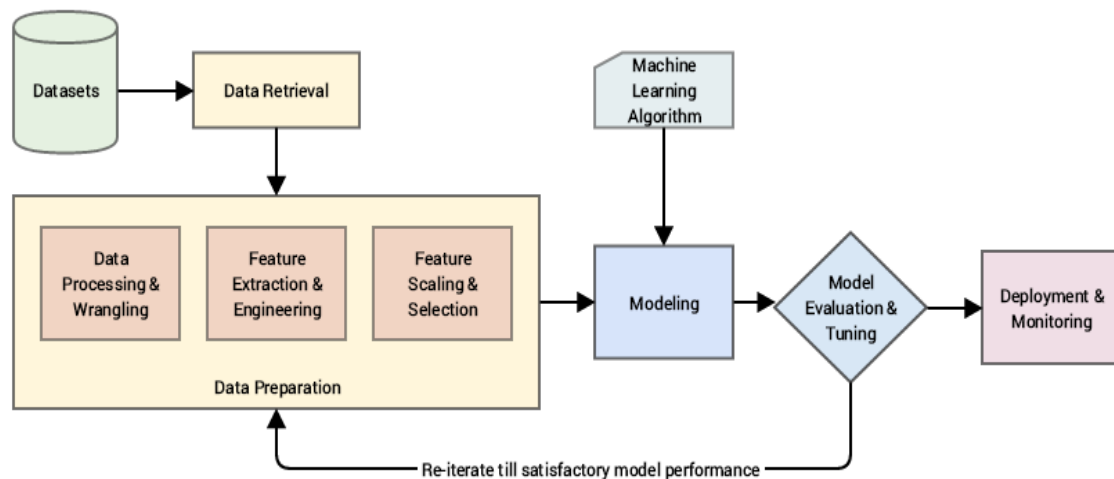


EDA Tasks:

- Check the target variable:
- Amount of attrition.
- Check the demographic variables:
- Age vs attrition.
- Gender vs churn.
- Number of dependents vs churn.
- Education level vs churn.
- Marital status vs churn.
- Income category vs churn.
- Check the product variables:
- Type of card vs churn.
- Relationship with the bank vs churn.
- Number of products vs churn.
- Inactive months vs churn.
- Number of contacts vs churn.
- Credit Limit vs churn.
- Total revolving balance vs churn.
- Openness To Buy Credit Line vs churn.
- Transaction Amount Change vs churn.
- Transaction Count Change vs churn.
- Average Card Utilization Ratio vs churn.

FEATURE ENGINEERING

Feature engineering refers to the process of using domain knowledge to select and transform the most relevant variables from raw data when creating a predictive model using machine learning or statistical modeling. The goal of feature engineering and selection is to improve the performance of machine learning (ML) algorithms.



Feature engineering is a machine learning technique that leverages data to create new variables that aren't in the training set. It can produce new features for both supervised and unsupervised learning, with the goal of simplifying and speeding up data transformations while also enhancing model accuracy. Feature engineering is required when working with machine learning models. Regardless of the data or architecture, a terrible feature will have a direct impact on your model.

It is the process of transforming raw data into features that are suitable for machine learning models. In other words, it is the process of selecting, extracting, and transforming the most relevant features from the available data to build more accurate and efficient machine learning models.

Importance Of Feature Engineering

Feature Engineering is a very important step in machine learning. Feature engineering refers to the process of designing artificial features into an algorithm. These artificial features are then used by that algorithm in order to improve its performance, or in other words reap better results. Data scientists spend most of their time with data, and it becomes important to make models accurate.

Steps In Feature Engineering

The art of feature engineering may vary among data scientists, however steps for how to perform feature engineering for most machine learning algorithms include the following:

1.Data Preparation:

This preprocessing step involves the manipulation and consolidation of raw data from different sources into a standardized format so that it can be used in a model. Data preparation may entail data augmentation, cleaning, delivery, fusion, ingestion, and/or loading.

2.Exploratory Analysis:

This step is used to identify and summarize the main characteristics in a data set through data analysis and investigation. Data science experts use data visualizations to better understand how best to manipulate data sources, to determine which statistical techniques are most appropriate for data analysis, and for choosing the right features for a model.

3.Benchmark:

Benchmarking is setting a baseline standard for accuracy to which all variables are compared. This is done to reduce the rate of error and improve a model's predictability. Experimentation, testing and optimizing metrics for benchmarking is performed by data scientists with domain expertise and business users.

ADVANCED REGRESSION TECHNIQUES

When dealing with advanced regression techniques like ensemble models and feature engineering for a customer churn prediction project, you can significantly enhance the predictive accuracy and robustness of your model. Here's how to incorporate these techniques:

1.Ensemble Models

a. Random Forest Ensemble

Implement a Random Forest Classifier to leverage the power of multiple decision trees. It's robust against overfitting and captures complex relationships.

b. Gradient Boosting Ensemble(e.g., XGBoost, LightGBM, CatBoost)

These algorithms create an ensemble of decision trees and optimize them sequentially. They often provide state-of-the-art performance in classification tasks like churn prediction.

c. Stacking Ensemble

Combine predictions from multiple models (e.g., logistic regression, random forest, gradient boosting) to create a meta-model. This meta-model can offer improved predictive performance.

2. Support Vector Machine

A Support Vector Machine is a supervised machine learning algorithm used for classification and regression tasks. It works by finding the optimal hyperplane that best separates data points into different classes. The key concept is SVM is to maximize the margin which is the distance between the hyperplane and the nearest data points.

3. Nearest Neighbour Classifier

A Nearest Neighbor Classifier also known as the k-Nearest Neighbor classifier, is a simple and intuitive machine learning algorithm used for classification and regression tasks. It operates based on the principle of proximity: objects that are close to each other in a feature space are likely to belong to the same class.

4. Evaluation Metrics

Choose appropriate evaluation metrics for your churn prediction task, such as accuracy, precision, recall, F1-score, ROC AUC, or customer lifetime value (CLV) metrics, depending on your business objectives.

5. Model Interpretability

Consider techniques like SHAP values or Partial Dependency Plots to interpret and explain the predictions of your ensemble models, especially if interpretability is crucial for decision-making.

6. Regularization Techniques

Apply regularization methods like L1 (Lasso) or L2 (Ridge) regularization to prevent overfitting in your models.

By combining ensemble techniques with effective feature engineering and thorough model evaluation, we can create a robust and accurate customer churn prediction model that will help your business proactively retain valuable customers.

MODEL EVALUATION AND SELECTION

Selecting and evaluating a model for a customer churn prediction project involves several steps. Here's a high-level overview:

1.Data Preparation:

- Collect and clean your data, handling missing values and outliers.
- Feature engineering: Create relevant features that might improve model performance.
- Split your data into training, validation, and test sets.

2. Selecting Algorithms:

- Choose a set of algorithms to evaluate.
- Common choices include logistic regression, decision trees, random forests, support vector machines, and gradient boosting methods like XGBoost or LightGBM.

3. Evaluation Metrics:

- Define appropriate evaluation metrics for your problem.
- For churn prediction, commonly used metrics include accuracy, precision, recall, F1-score, ROC AUC, and lift.

4. Baseline Model:

- Start with a simple baseline model.
- It can be a logistic regression model or a decision tree, which provides a benchmark for more complex models.

5. Model Training:

- Train each selected model using the training dataset.
- Tune hyperparameters using techniques like grid search or random search.

6. Model Evaluation:

- Evaluate each model's performance on the validation dataset using the chosen metrics.
- Consider model interpretability for business insights.

MODEL INTERPRETABILITY

The demand for interpretability increases when there is a large cost for incorrect predictions, especially in high-risk applications. Based on the model complexity, methods for model interpretability can be classified into intrinsic analysis and post hoc analysis.

The model itself becomes the source of knowledge instead of the data. Interpretability makes it possible to extract this additional knowledge captured by the model. Machine learning models take on real-world tasks that require safety measures and testing. It can help improve interpretability by reducing the model's complexity and avoiding unnecessary features. You can use regularization techniques such as Lasso, Ridge, or Elastic Net to shrink or eliminate the coefficients of less important features, making the model more sparse and simple.

DEPLOYMENT AND PREDICTION

Here we will train an optimized (treebased) model which will predict if a customer will or won't churn.

Data Preparation

Before we start training a model we must prepare our data. Different steps that we can undertake:

- Encode all categorical data (watch out with one hot encoding and tree-based models...).
- Scale data
- Check correlation matrix to extract the most influential features.
- Generate new columns from data.
- Upsample the imbalanced dataset (SMOTE/ADASYN).

In this notebook we shall focus on the upsampling method. The data wrangling performed is to make sure that the upsampling is performed in a correct manner.

SMOTE (Synthetic Minority Oversampling Technique)

We saw that our dataset was imbalanced. This could give problems when creating a classification model since it might not learn the decision boundary. This of course can be solved with upsampling.

One technique used for this is SMOTE, this technique creates new synthetic samples which can be used for training.

SMOTE first selects a minority class instance a at random and finds its k nearest minority class neighbors. The synthetic instance is then created by choosing one of the k nearest neighbors b at random and connecting a and b to form a line segment in the feature space. The

synthetic instances are generated as a convex combination of the two chosen instances a and b.

SMOTE Paper

To use SMOTE we'll need to encode our categorical features.

Note: It's important to only upsample the training data and so that no synthetic data is present in the validation dataset

PROGRAM

Customer Churn Prediction

Importing Dependencies

In[1] :

```
From mpl_toolkits.mplot3d import Axes3D
```

```
From sklearn.preprocessing import StandardScaler
```

```
Import matplotlib.pyplot as plt
```

```
Import numpy as np
```

```
Import os
```

```
Import pandas as pd
```

```
Print(os.listdir(['WA_Fn-UseC_-Telco-Customer-Churn.csv']))
```

Model 1 - Logistic Regression

In[2]:

```
Def logistic_regression(frame, grid):
```

```
    Logit = Model(LogisticRegression, frame, metrics)
```

```
    Print_result(logit.grid_search('roc_auc', grid), "cross-validation")
```

```
    Print_result(logit.train_and_evaluate_test(), "test set")
```

```
Print("Best hyperparameters:", logit.hyperparameters)
```

```
Logistic_regression(df_processed, {'C': np.logspace(-4, 4, 100, base=10)})
```

Out[2]:

```
{'random_state': 50}
```

```
| roc_auc: 0.8463 | accuracy: 0.8046 – cross-validation
```

```
| roc_auc: 0.8399 | accuracy: 0.7999 – test set
```

```
Best hyperparameters: {'C': 1.3219411484660315}
```

In[3]:

```
Logit_grid = {'C': np.linspace(0.02, 3, 150)}
```

```
Logistic_regression(df_processed, logit_grid)
```

Out[3]:

```
{'random_state': 50}
```

```
| roc_auc: 0.8463 | accuracy: 0.8046 – cross-validation
```

```
| roc_auc: 0.8399 | accuracy: 0.7999 – test set
```

```
Best hyperparameters: {'C': 1.3800000000000001}
```

Model 2 - Support Vector Machine

Rbf Kernel

In[4]:

```
Def svc_rbf(frame, grid):
```

```
    Rbf = Model(SVC, frame, metrics, fixed_params= {'kernel': 'rbf', 'probability': True})
```

```

Print_result(rbf.grid_search('roc_auc', grid, num_folds=4), "cross-validation")

Print_result(rbf.train_and_evaluate_test(), "test set")

Print("Best hyperparameters:", rbf.hyperparameters)

Grid_rbf = {'C': np.logspace(-4, 1, 10, base=2), 'gamma': np.logspace(-6, 2, 10, base=2)}

Svc_rbf(df_processed, grid_rbf)

```

Out[4]:

```

{'kernel': 'rbf', 'probability': True, 'random_state': 50}

| roc_auc: 0.8365 | accuracy: 0.7969 – cross-validation

| roc_auc: 0.8352 | accuracy: 0.7999 – test set

Best hyperparameters: {'C': 0.13500746736153826, 'gamma': 0.015625}

```

Model 3 - Nearest Neighbors Classifier

In[5]:

```

Def knn_clf(frame, grid):

    Knn = Model(KNeighborsClassifier, frame, metrics)

    Print_result(knn.grid_search('roc_auc', grid), "cross-validation")

    Print_result(knn.train_and_evaluate_test(), "test set")

    Print("Best hyperparameters:", knn.hyperparameters)

Knn_clf(df_processed, {'n_neighbors': [i for i in range(10, 50, 2)]})

```

Out[5]:

```

{}

| roc_auc: 0.8360 | accuracy: 0.7984 – cross-validation

```

| roc_auc: 0.8324 | accuracy: 0.7906 – test set

Best hyperparameters: {'n_neighbors': 48}

Model 4 - Feature Engineering

Removing Features

In[6]:

```
# Remove Gender
```

```
Features = ['gender']
```

```
Df_processed.drop(features, axis=1, inplace=True)
```

```
Logit = Model(LogisticRegression, df_processed, metrics)
```

```
Print_result(logit.grid_search('roc_auc', logit_grid), "cross-validation")
```

Out[6]:

```
{'random_state': 50}
```

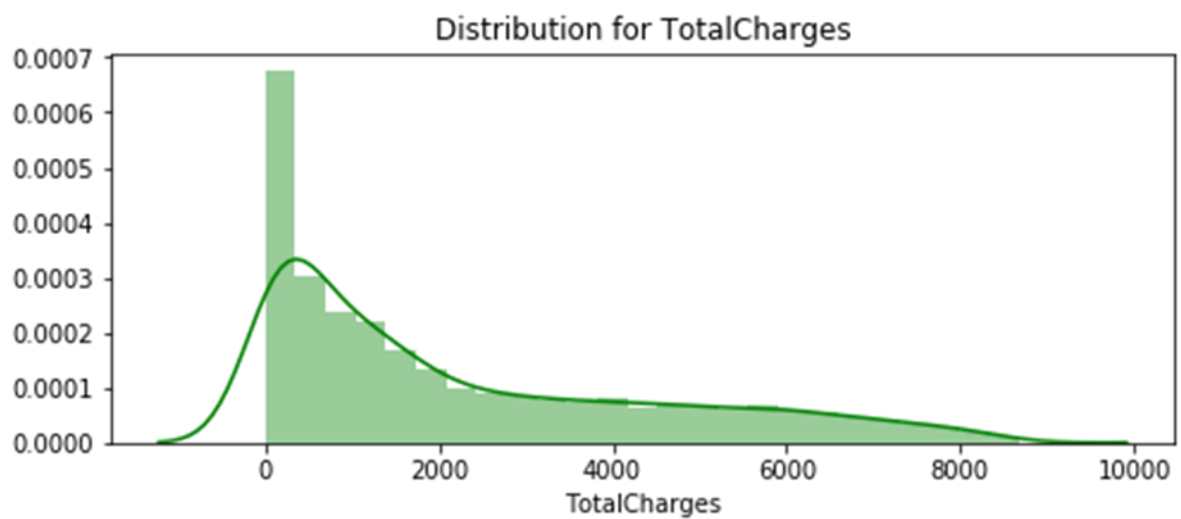
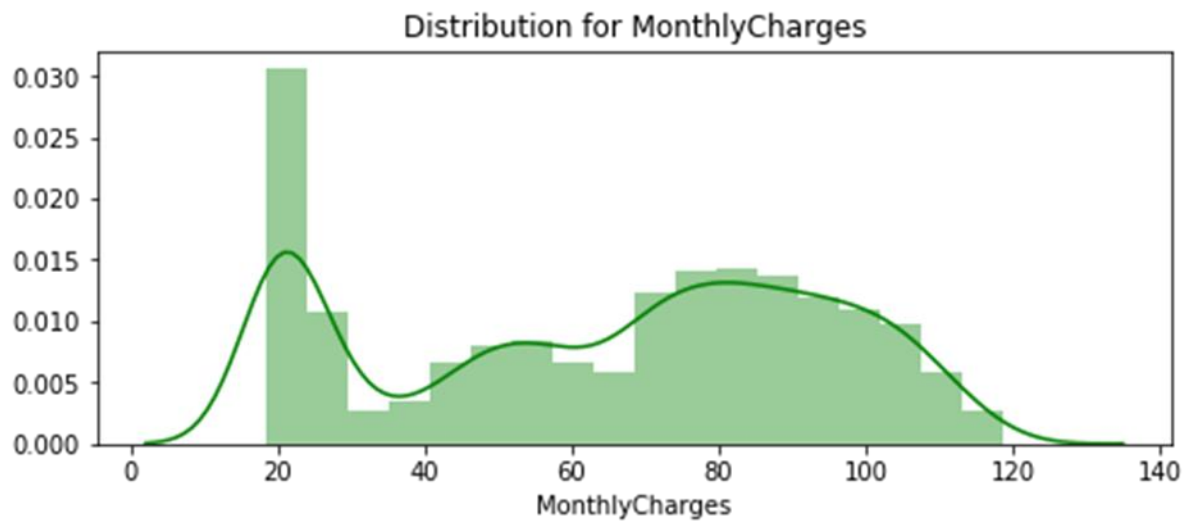
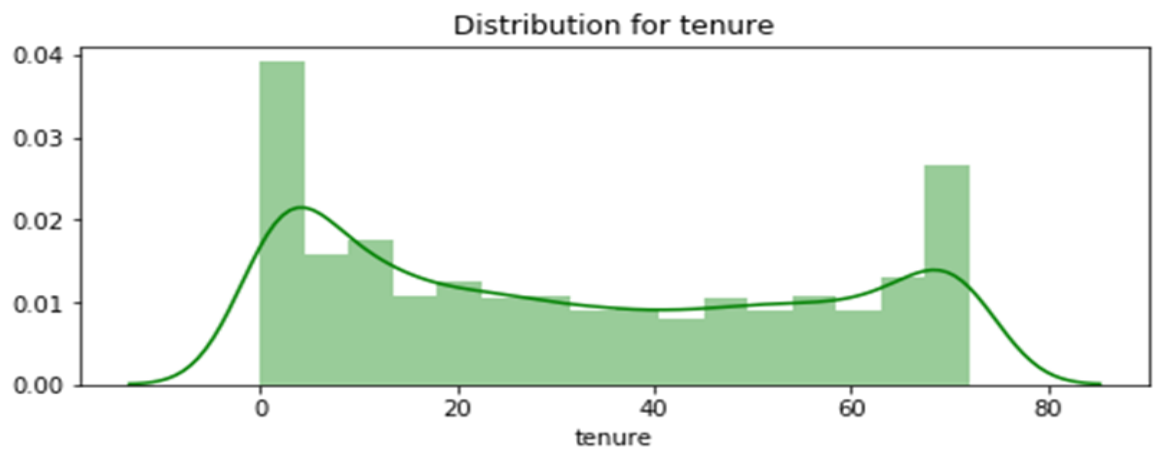
| roc_auc: 0.8463 | accuracy: 0.8051 – cross-validation

Feature Scalling

In[7]:

```
For feat in numerical_cols: distplot(feat, df)
```

Out[7]:



CONCLUSION

In summary, the customer churn prediction project has not only reduced customer churn but also positioned the business for long-term success in a competitive market. By embracing data-driven approaches, we are better equipped to meet customer needs, maximize customer satisfaction, and drive sustainable business growth. This project underscores the importance of proactive customer retention strategies and serves as a model for businesses seeking to leverage data analytics to optimize customer relationships

Customer churn analysis allows to minimize acquisition costs and increase marketing efficiency, preparing a solid base for future marketing analysis and campaigns. Customer churn analysis opens new opportunities for cross-selling and upselling and serves as one of the starting points for customer-driven product development, keeping customers engaged and loyal over time.